

## CHEMICAL ANALYSIS – TESTS FOR ANIONS

## KEY VOCABULARY

**Carbonate test:** add acid → fizzes, CO<sub>2</sub> turns limewater cloudy.

**Halide test:** add nitric acid then silver nitrate → coloured precipitate.

**Sulfate test:** add hydrochloric acid then barium chloride → white precipitate.

**Anion:** a negative ion being tested for.

## COMMON MISCONCEPTIONS

~~All silver halides are white.~~

✓ AgCl white, AgBr cream, AgI yellow.

~~Add barium chloride alone for sulfates.~~

✓ Add dilute HCl first to remove carbonates.

~~Carbonates need silver nitrate.~~

✓ Carbonates are tested with dilute acid + limewater.

## ANION TEST SUMMARY

**Carbonate:** + dilute acid → fizz (CO<sub>2</sub>).

**Sulfate:** + HCl + BaCl<sub>2</sub> → white precipitate.

**Halide:** + HNO<sub>3</sub> + AgNO<sub>3</sub> → precipitate.

Cl<sup>-</sup> white · Br<sup>-</sup> cream · I<sup>-</sup> yellow.

Add the acid first to remove other ions.

## 1 CARBONATE

Describe the test and result for a **carbonate**. [2 marks]

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## 2 SULFATE

Describe the test and result for a **sulfate**. [2 marks]

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## 3 HALIDES

State the precipitate colours for chloride, bromide and iodide with silver nitrate. [3 marks]

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## 4 WHY ADD ACID

Explain why dilute acid is added before the main test reagent. [2 marks]

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## 5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A sulfate gives a \_\_\_\_ precipitate with \_\_\_\_ chloride.

white

cream

yellow

barium

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## 6 IDENTIFY

A solution gives a cream precipitate with AgNO<sub>3</sub>. **Identify** the ion. [1 mark]

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